

Agora Scientific Document

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A Monograph on the Phase 2 Results of the HorizonMath Initiative

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Abstract

The HorizonMath initiative, a core component of the SymBrain v16 project, aims to build a vast, formally verified library of advanced mathematics. This monograph details the complete results of Phase 2, which focused on the formalization and preliminary verification of 50 foundational theorems across diverse mathematical domains. A key aspect of this phase was the strategic use of admitted proofs (axioms or "sorries") to rapidly sketch the high-level structure of the library, deferring complex, low-level proofs to subsequent phases. We present the full list of 50 formalized statements, along with their sanitized proof-placeholders. This work establishes a robust scaffold for the automated and human-assisted proof generation efforts planned for Phase 3, marking a significant milestone in the SymBrain project's long-term vision.

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1 Introduction

The SymBrain project at Socrate AI Lab is dedicated to exploring the confluence of symbolic reasoning and neural computation. A cornerstone of this research is the HorizonMath initiative, which seeks to construct a comprehensive and machine-readable corpus of mathematics. The ultimate goal is to create a formal system that is not only verifiable but also amenable to exploration and discovery by advanced AI agents.

Phase 1 of HorizonMath focused on establishing the core logical framework and type theory. This monograph addresses the outcomes of Phase 2, whose primary objective was to map out the landscape of a significant portion of graduate-level mathematics. This was achieved by formalizing the statements of 50 key theorems and admitting their proofs. This "proof-sketching" methodology allows for rapid development of a dependency graph between major mathematical results, identifying critical lemmas and definitions required for full verification.

This document serves as the canonical reference for the 50 results of Phase 2. We provide:

- A summary of the methodology used for formalization and sanitization of admitted proofs.
- A comprehensive table listing all 50 theorem statements.
- A detailed appendix with the formal definitions and sanitized proof scripts for a representative subset of these theorems.

The successful completion of Phase 2 provides the necessary foundation for the targeted proof-infilling efforts of Phase 3, which will leverage both automated theorem provers and human-in-the-loop verification systems.

2 Methodology

The core of our Phase 2 strategy was the separation of theorem statement formalization from the proof-finding process. This allowed our team of mathematicians and formalization experts to focus on accurately translating complex mathematical ideas into our formal language without being immediately encumbered by the often immense effort required for a full, rigorous proof.

2.1 Formalization Framework

Our framework is based on a dependent type theory, similar in spirit to Lean and Coq. All definitions and theorems are checked for type-correctness by the kernel. The language is designed to be expressive, allowing for natural statements in fields ranging from abstract algebra to differential geometry.

2.2 Sanitized ‘sorry’ Proofs

In our system, the keyword ‘sorry’ is a tactic that closes any proof goal, effectively admitting the theorem as an axiom. While powerful for rapid prototyping, overuse of ‘sorry’ can lead to an inconsistent library. Our sanitization protocol requires that every use of ‘sorry’ is justified and documented. A sanitized proof must include:

1. The ‘sorry’ keyword, clearly marking the proof as incomplete.
2. A structured comment detailing the reason for admission (e.g., proof complexity, dependency on an unformalized theory).
3. An estimate of the proof effort and a reference to the relevant literature.

This protocol ensures that each admitted proof serves as a well-defined work item for Phase 3, transforming a potential weakness into a project management tool.

3 Summary of Phase 2 Results

The 50 theorems of Phase 2 were selected to cover a wide range of mathematical disciplines, testing the expressivity of our formal language and establishing a broad foundation for future work. The following table summarizes these results. The full formal statements are provided in the appendix.

Table 1: The 50 Formalized Theorems of HorizonMath Phase 2

ID	Informal Statement	Status
HM.P2.01	Atiyah-Singer Index Theorem for Dirac Operators	Admitted (Sanitized)
HM.P2.02	Existence of Nash Equilibria in Finite Games	Admitted (Sanitized)
HM.P2.03	Central Limit Theorem for i.i.d. Random Variables	Admitted (Sanitized)
HM.P2.04	Grothendieck-Riemann-Roch Theorem	Admitted (Sanitized)
HM.P2.05	Perelman’s Theorem on Poincaré Conjecture (for S^3)	Admitted (Sanitized)

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ID	Informal Statement	Status
HM.P2.06	Unique Fixed Point for SymBrain Operator $\mathcal{S}_\theta$	Admitted (Sanitized)
HM.P2.07	Regularity of Solutions to the Navier-Stokes Equations	Admitted (Sanitized)
HM.P2.08	Fermat’s Last Theorem	Admitted (Sanitized)
HM.P2.09	The Prime Number Theorem	Admitted (Sanitized)
HM.P2.10	Gödel’s First Incompleteness Theorem	Admitted (Sanitized)
HM.P2.11	Hales’ Theorem on Kepler Conjecture	Admitted (Sanitized)
HM.P2.12	Mordell-Weil Theorem for Elliptic Curves	Admitted (Sanitized)
HM.P2.13	Baire Category Theorem	Admitted (Sanitized)
HM.P2.14	Spectral Theorem for Compact Self-Adjoint Operators	Admitted (Sanitized)
HM.P2.15	Yoneda Lemma in Category Theory	Admitted (Sanitized)
HM.P2.16	Gauss-Bonnet Theorem	Admitted (Sanitized)
HM.P2.17	Van Kampen’s Theorem	Admitted (Sanitized)
HM.P2.18	Law of Large Numbers (Strong Version)	Admitted (Sanitized)
HM.P2.19	Isomorphism between Čech and Sheaf Cohomology	Admitted (Sanitized)
HM.P2.20	Zorn’s Lemma	Admitted (Sanitized)
HM.P2.21	Shannon’s Source Coding Theorem	Admitted (Sanitized)
HM.P2.22	Fast Fourier Transform (FFT) Correctness	Admitted (Sanitized)
HM.P2.23	Cook-Levin Theorem (SAT is NP-complete)	Admitted (Sanitized)
HM.P2.24	Hölder’s Inequality	Admitted (Sanitized)
HM.P2.25	Cauchy’s Integral Formula	Admitted (Sanitized)
...	...	...
HM.P2.50	The Universal Approximation Theorem for Neural Networks	Admitted (Sanitized)

4 Discussion and Future Work

The completion of Phase 2 represents a paradigm shift in the HorizonMath initiative. We have moved from foundational logic-building to broad mathematical cartography. The 50 results listed in Table ?? now form a network of interconnected goals for our proof generation systems.

The process of formalizing these diverse statements has been invaluable. It stress-tested our type theory, leading to several improvements in the ergonomics of defining complex structures like schemes, manifolds, and probability spaces. The sanitization protocol for admitted proofs has proven effective, creating a clear and actionable roadmap for Phase 3.

Phase 3 will commence immediately and will involve a two-pronged approach:

1. **Automated Proving:** A suite of automated theorem provers, including SMT solvers and reinforcement learning-based tactic selectors, will be deployed to attempt to discharge the ‘sorry’ obligations for theorems with known, highly syntactic proofs.

2. **Human-AI Collaboration:** For conceptually deeper proofs, a collaborative environment will allow human mathematicians to guide AI assistants, breaking down complex problems into manageable lemmas. The system will then attempt to prove these lemmas automatically.

Our goal for the next cycle is to achieve full verification for at least 20% of the admitted theorems and to substantially reduce the complexity of another 30% by proving key intermediate lemmas.

5 Conclusion

The HorizonMath Phase 2 has successfully established a vast, structured, and formally-defined frontier for mathematical exploration by AI. By formalizing 50 of the most significant theorems in mathematics and embedding them in a rigorous framework, we have created a rich set of challenges and a clear path forward. The sanitized proofs, far from being a liability, are the very blueprint for the next stage of automated and semi-automated mathematical discovery. This monograph stands as the definitive record of this achievement and the foundational document for the exciting work that lies ahead in Phase 3 of the SymBrain project.

A Formal Statements and Sanitized Proofs

This appendix contains the formal code for a representative selection of the 50 theorems from Phase 2. Each entry includes the formal statement in our system’s syntax and the sanitized proof block.

A.1 HM.P2.01: Atiyah-Singer Index Theorem

Theorem A.1 (Atiyah-Singer Index Theorem). *Let M be a compact, oriented smooth manifold, and let D be a Dirac operator on a complex vector bundle $E \rightarrow M$. Then the analytical index of D is equal to its topological index.*

```

1 theorem atiyah_singer_index_theorem
2   (M : CompactOrientedSmoothManifold)
3   (E : VectorBundle C M)
4   (D : DiracOperator E) :
5   anal_index(D) = topo_index(D) :=
6 begin
7   sorry,
8   --[[
9   Sanitization Report:
10  - Reason: Proof requires the full machinery of K-theory and the theory
11    of elliptic pseudodifferential operators, neither of which is
12    fully formalized yet.
13  - Dependencies: Formalization of Chern character, Todd class, and
14    the Thom isomorphism.
15  - Literature: Atiyah, M. F., & Singer, I. M. (1968). The index of
16    elliptic operators on compact manifolds.
17  - Est. Effort: > 10 person-years.
18  --]]
19 end

```

Listing 1: Sanitized Proof for `atiyah_singer_index_theorem`

A.2 HM.P2.02: Nash Equilibrium Existence

Theorem A.2 (Nash Equilibrium Existence). *Every finite game with n players has at least one Nash equilibrium in mixed strategies.*

```
1 theorem nash_equilibrium_existence
2   (G : FiniteGame) :
3   exists (s : MixedStrategyProfile G), is_nash_equilibrium G s :=
4 begin
5   sorry,
6   --[[
7   Sanitization Report:
8   - Reason: Proof relies on the Brouwer fixed-point theorem. While the
9     statement of Brouwer's theorem is formalized (HM.P2.35), its proof
10    is also admitted. This proof will be unblocked once Brouwer's theorem
11    is fully proven.
12    - Dependencies: `brouwer_fixed_point_theorem`.
13    - Literature: Nash, J. (1951). Non-cooperative games.
14    - Est. Effort: 3-4 person-months post-Brouwer.
15    --]]
16 end
```

Listing 2: Sanitized Proof for `nash_equilibrium_existence`

A.3 HM.P2.06: SymBrain Operator Fixed Point

Definition A.3 (SymBrain Operator). A SymBrain Operator $\mathcal{S}_\theta$ is a map on a Hilbert space of cognitive states $\mathcal{H}$ parameterized by $\theta \in \Theta$.

Theorem A.4 (Unique Fixed Point). *For any well-posed SymBrain Operator $\mathcal{S}_\theta$ acting on a complete metric space $\mathcal{H}$, there exists a unique fixed point $h^* \in \mathcal{H}$ such that $\mathcal{S}_\theta(h^*) = h^*$.*

```
1 theorem symbrain_operator_fixed_point
2   (H : HilbertSpace)
3   (theta : ParameterSpace)
4   (S : SymBrainOperator H theta)
5   (h_is_complete : IsComplete H)
6   (s_is_contraction : IsContraction S) :
7   exists! (h_star : H), S(h_star) = h_star :=
8 begin
9   sorry,
10  --[[
11  Sanitization Report:
12  - Reason: The proof follows directly from the Banach fixed-point theorem.
13    The goal here is to formalize the specific conditions under which a
14    SymBrain operator is a contraction mapping. This requires further
15    modeling of the operator's properties.
16    - Dependencies: `banach_fixed_point_theorem`.
17    - Literature: Internal Socrate AI Lab theoretical documentation.
18    - Est. Effort: 2 person-months.
19    --]]
20 end
```

Listing 3: Sanitized Proof for `symbrain_operator_fixed_point`